

Chapter 148 — Anatomy and Development of the Prostate

PROSTATE ANATOMY

Zonal Anatomy of the Human Prostate

- **Fetal origin:** epithelial cords from the primitive urethra canalise and branch in response to **androgen action in local mesenchyme**.
- **McNeal zones (1981):** peripheral zone (PZ), central zone (CZ), preprostatic region (transition zone, TZ) and anterior fibromuscular stroma (AFMS).
- Zones were defined partly by **where focal disease arises:** benign prostatic hyperplasia (BPH) and cancer (table).
- **Orientation axes:** base (bladder neck) → apex (junction of prostatic and bulbar urethra); anterior (top) → dorsal (bottom).
- The **prostatic urethra bends approximately 35°** before the dorsal entry of the ejaculatory and prostate ducts.
- **13–15 prostate ducts** open onto the dorsolateral prostatic urethra along the base-to-apex axis, best shown by dorsal nano computed tomography (nanoCT).
- **Verumontanum:** a bulge distal to the urethral crest that marks the entry of the ejaculatory ducts (seminal vesicle and vas deferens).

ZONES OF THE HUMAN PROSTATE (MCNEAL)

Zone	Defining ductal anatomy	Disease association
Transition zone	Shortest ducts; terminal secretory glands most proximal to the urethra	BPH
Peripheral zone	Distal glands with the longest ducts	70% of cancers
Central zone	Ducts and glands running parallel to the baso-dorsal entry of the ejaculatory ducts	—
AFMS	Continuous ridge of fibroblasts and smooth muscle along the anterior bladder neck and prostate	—

- The table tests **duct length as the zonal discriminator:** short proximal ducts form the TZ and long distal ducts form the PZ. The trap is to confuse the CZ, which surrounds the ejaculatory ducts, with the TZ.

Concept — zones are defined by **duct length and position relative to the urethra and ejaculatory ducts**, and the disease map follows: proximal short-duct tissue enlarges (BPH), while distal long-duct tissue transforms (cancer).

Cellular Anatomy of Human Prostate

- Cell types, once defined by morphology, location and selective protein or RNA expression, are now defined by **pan-transcriptomic profile** (single-cell or single-nucleus RNA sequencing) plus in situ location and morphology; modern prostate anatomy is the **distribution of these discrete cell types**, divided into epithelia and stroma.
- Validated types are annotated in the **Cell Ontology** (a structured vocabulary for animal cell types) and placed anatomically in **Uberon** (a cross-species anatomy ontology). The **Human BioMolecular Atlas**

Program (HuBMAP) and Human Cell Atlas aim to build three-dimensional tissue maps and an atlas of cell function and relationships.

- **Acinus:** pseudostratified columnar epithelium (basal and luminal cells) immediately surrounded by periepithelial fibroblasts; interacinar PZ stroma holds interstitial fibroblasts and smooth muscle. Gland plus stroma = **functional tissue unit**.

CELL TYPES OF THE HUMAN PROSTATE AND PROSTATIC URETHRA

Compartment	Cell type	Markers
Prostate epithelium	Luminal	KLK3+/NKX3-1+
	Basal	KRT14+/KRT5+
Urethral and proximal-duct epithelium	Club	SCGB3A1+/SCGB1A1+
	Hillock	KRT13+/LYPD3+
	Neuroendocrine	CHGA+/ASCL1+
	Ionocyte	FOXI1+/RCAN2+
Stroma	Periepithelial fibroblast	APOD+/PTGDS+
	Interstitial fibroblast	C7+/MFAP4+
	Prostate smooth muscle	DES+/ACTG2+
Endothelium	Arterial / capillary / venous / lymphatic	GJA4+/IGFBP3+ · GJA1+/CA4+ · SELP+/CSF3+ · LYVE1+/FLT4+
Mural cells	Vascular smooth muscle	MCAM+/ACTA2+
	Pericyte	CD36+/ACTA2+
Immune	Resident immune cells	—

- The table tests **marker-to-cell matching**. The pattern is that the urethra and proximal ducts carry an **airway-like epithelium** (club, hillock, ionocyte, neuroendocrine), while NKX3-1 identifies true secretory prostate epithelium.

Luminal Epithelia of Prostate

- The "**workhorse**" of the gland, responsible for epithelial barrier integrity and protein secretion.
- Their products, important for sperm function, include **KLK3 (kallikrein-related peptidase 3, prostate-specific antigen, PSA)**, **prostatic acid phosphatase (PAP)** and **androgen receptor (AR)**.
- **Polarity:** rows joined by adhesion molecules; base anchored to basement membrane by integrins; basal nucleus beneath a Golgi-rich clear zone (2–8 µm); apex with microvilli and abundant secretory granules. Secretions pass acinus → ducts → urethra.

Basal Epithelia of Prostate

- Basal cells have **condensed chromatin** and rest on the basement membrane, **wedged between the bases of the luminal cells**. Their progenitor role is limited to certain situations; their main function is ductal integrity.

LUMINAL VERSUS BASAL EPITHELIUM

Feature	Luminal	Basal
Keratins	8 and 18	5 and 14
Shape	Tall columnar, 10–20 µm	Pyramidal, scant cytoplasm
AR	Expressed	Low

Feature	Luminal	Basal
Secretion	KLK3/PSA, PAP	Almost none
Proliferation / fate	Terminally differentiated, low proliferative index	Can act as luminal progenitor
Share of epithelium	Majority	Minority
Role	Barrier and secretion	Ductal integrity

Club and Hillock Epithelia of Prostatic Urethra

- The prostatic urethra is lined by a **transitional epithelium containing club and hillock cells**, like those of the proximal lung.
- Club and hillock cells extend into the proximal ducts before they transition to NKX3-1+ prostate epithelium. This gives a **cell type-specific definition of the "transition zone"**.
- Lung club cells detoxify, regulate immunity and regenerate several cell types after injury; a similar prostatic role is presumed but **untestable in the mouse, which lacks club cells**.

Neuroendocrine and Ionocyte Epithelia of Prostate and Prostatic Urethra

- Like the proximal lung, the prostatic urethra contains **neuroendocrine cells and ionocytes**. Ionocytes are thought to **break down seminal fluid**, as they break down mucus in the lung.
- Neuroendocrine cells are **chemo- and mechanosensory**; in lung and intestine they release hormones and antimicrobial peptides under stress and interact with nervous, endocrine and immune systems. **Their prostatic function is poorly characterised**.
- Their presence in **male and female urogenital sinus epithelium before prostate development** suggests a lineage independent of prostate epithelium; lineage tracing suggests **most are neural crest-derived**. What triggers epithelial transdifferentiation or neural crest migration is undefined.

Prostate Fibroblasts and Smooth Muscle

- A **dense population of interstitial fibroblasts surrounds the prostatic urethra** and its ducts, presumably providing structural support during voiding and ejaculation.
- Periepithelial and interstitial fibroblasts were **discovered only recently**. Transcriptomic signatures suggest roles in **stress response and immune response** respectively.

Prostate Vasculature

- Vascular smooth muscle (large vessels) and pericytes (microvessels) are both **mural cells**: they are embedded in basement membrane and extend long processes that interact with and regulate endothelium.

Prostate Immunology

- Healthy young donor prostate (single-cell RNA sequencing) has a **small resident population of macrophages, dendritic cells and T cells**, also found in adult mouse prostate.
- BPH tissue has prominent **T cells, B cells and dendritic cells** and resembles a **mucosal inflammatory disorder**.

Figure legend

- **Fig. 148.3 (Joseph et al., Prostate, 2020)**: the markers are **keratin 5 for basal cells** (mouse and human), **keratin 13 for hillock cells** of the urethra and proximal ducts (human), **Krt4 for luminal cells of the mouse**

urethra and proximal ducts, and NKX3-1 for secretory luminal cells (both species). The source identifies **urethral luminal epithelia as castration-insensitive cells of the proximal prostate**.

Concept — the prostate is now mapped by **cell identity rather than landmarks**. The urethra and proximal ducts carry a lung-like epithelium, and the "transition zone" becomes the point at which **urethral epithelium hands over to NKX3-1+ prostate epithelium**.

Lobar Anatomy of Murine Prostate

- The mouse lobes are **named for their urethral site of origin**; each has a unique branching pattern, histology and transcriptionally distinct luminal epithelium, and lobes have been likened to human zones.

HUMAN VERSUS MOUSE PROSTATE

Feature	Human	Mouse
Architecture	Encapsulated, zonal	Lobar (anterior, ventral, dorsolateral)
PZ correlate	Peripheral zone	Dorsolateral lobe
PSA (KLK3)	Present; secretory biomarker	Absent from the genome ; other kallikreins expressed
Club cells	Present in urethra	Absent
Luminal promoter used experimentally	—	Probasin : expressed in all lobes' luminal cells; drives selective transgene expression

Concept — the mouse reproduces **developmental mechanism but not human anatomy**. It lacks a capsule, zones, PSA and club cells, so findings on these features cannot be transferred directly.

Prostatic Secretions

- The prostate's overarching function is to **synthesise and secrete factors that support the safe passage of sperm** from the urethra to the egg.
- Semen contains cells, salts, sugars, hormones, carbohydrates, lipids and **more than 2000 proteins**; some prostate-selective factors serve **diagnosis, imaging or treatment**.

PROSTATIC VERSUS SEMINAL VESICLE FLUID

Feature	Prostatic fluid	Seminal vesicle fluid
Share of semen volume	37–44%	55–61%
pH	Acidic (about 6.3)	Basic
Key contents	Kallikreins (PSA, KLK2), PAP, zinc, citrate, polyamines	Semenogelins (coagulum substrate)

- Combined, the acidic and basic fluids give semen **considerable buffering capacity**.

Ion trapping and drug distribution

- Prostatic fluid (pH about 6.3) is more acidic than plasma (7.4). **Un-ionised drugs diffuse**; ionised drugs need transport, so a drug **accumulates where it is most ionised** — one reason **macrolides concentrate in prostatic fluid**.
- Inflammation alkalinises prostatic fluid (above pH 8.3)**, and pH also **rises with age**; both alter intraprostatic drug distribution.

Coagulation and liquefaction

- Sequence**: prostatic and seminal vesicle fluids mix → **coagulation** (a gel entraps and immobilises sperm during ejaculation) → **liquefaction** (kallikreins and other proteases degrade semenogelins) → motile sperm.

- **Normal liquefaction: within 20–30 minutes.** Delayed if not begun by 30 minutes; **nonliquefaction** if not complete by 60 minutes.
- **Hyperviscosity is a suspected cause of male infertility.** Kallikrein inhibition to block liquefaction is under study as **male contraception**.

Kallikreins

- Kallikreins are among the **most abundant proteins in semen**. They form the **largest continuous protease cluster in the human genome (chromosome 19q13.4)**, which encodes **15** structurally similar proteins.
- Made as **inactive propeptides**, activated by extracellular autocatalytic or protease cleavage; regulated by **zinc, citrate, sulfate and glycosaminoglycans**.
- **PSA (KLK3)** is the kallikrein most associated with the prostate; its **mean whole-ejaculate concentration is about 1.2 mg/mL**. It is made by benign and malignant prostate epithelium and by several extraprostatic tissues (table below).
- **KLK2** is also highly expressed in the prostate. It is related in sequence to KLK3 but has **distinct protease activity**.

Prostate-specific membrane antigen, prostatic acid phosphatase and polyamines

PROSTATE SECRETORY AND MEMBRANE PRODUCTS

Product	Gene locus	Nature	Extraprostatic sites	Function	Clinical relevance
PSA (KLK3)	19q13.4 cluster	Secreted serine protease	Skene gland, breast, amniotic fluid, ovarian carcinoma	Liquefaction	Serum biomarker
PSMA (folate hydrolase)	11p (rarely deleted in cancer)	Membrane-spanning carboxypeptidase	Heart, kidney, intestine, nervous system	Gut: dietary folate absorption; brain: N-acetyl-L-aspartyl-L-glutamate metabolism (glutamate-receptor control); prostate: polyglutamated folates → folic acid for trophic support (hypothesised)	PSMA-PET; ¹⁷⁷Lu-PSMA radioligand therapy
PAP (acid phosphatase 3)	Chromosome 3	Hydrolyses phosphoesters in acid conditions; intracellular and secreted forms	Brain, kidney, liver, lung, muscle, placenta, salivary gland, spleen, thyroid, thymus	Intracellular: controls phosphorylation (target ERBB2); secreted: seminal (1–2 mg/mL), enhances HIV infection	Pre-PSA tumour-burden marker (Huggins and Hodges, 1941)
Polyamines (spermine, spermidine, putrescine)	—	Polycationic; two or more amino groups	Ubiquitous; also dietary and microbial	Protect sperm from oxidative stress; increase motility	Lower in infertile men

- The table tests "**prostate-specific**" as a **misnomer**: PSA, PSMA and PAP are all expressed outside the prostate.
- **PSMA** is one of the most prominent membrane proteins of prostate cancer and **its neovasculature**, and its locus is rarely deleted, making it a **durable marker and target**. PSMA-PET shows promise over bone

scan, computed tomography (CT) and magnetic resonance imaging (MRI) in locally advanced and metastatic disease; ^{177}Lu -PSMA in advanced metastatic disease.

- **PAP forms arise by post-transcriptional modification.**
- **Seminal polyamines are more abundant than in any other body fluid** and are synthesised in prostate epithelium by **ornithine decarboxylase under androgen control.**

Zinc

- The prostate has the **highest zinc concentration of any human tissue**. The adult median is about **109 mg/kg**, with wide variation, and the concentration **rises with age**. **Seminal zinc (0.5 mg/mL) far exceeds plasma zinc (2 µg/mL).**
- **Transporters: ZIP proteins** (Zrt-, Irt-like proteins; solute carrier 39, SLC39) carry zinc **into** epithelial cells from plasma or prostatic fluid; **ZnT proteins** (zinc transporters; SLC30) carry zinc **out** into prostatic fluid.
- **Seminal zinc supports fertility** (low levels → infertility); it reduces oxidative stress, prevents citrate oxidation, **inhibits liquefaction** and correlates with motility.
- Zinc **inhibits the sperm voltage-gated proton channel (Hv1)**, blocking **capacitation** — the changes a sperm needs to penetrate and fertilise the egg — until diluted in the female tract.

Citrate and energy metabolism

- In most cells, citrate is an intermediate of glucose oxidation that is **oxidised through the Krebs cycle** to adenosine triphosphate (ATP). **Mitochondrial aconitase** converts citrate to isocitrate.
- In luminal cells **high zinc inhibits aconitase**, so citrate is stored and secreted, and benign cells derive energy from **oxaloacetate imported through a transmembrane transporter** — a **unique, inefficient strategy**.
- Citrate exceeds **100 mM** in the expressed prostatic secretions of most men without cancer, reaching up to **200 mM**, compared with **0.1 mM in plasma**. Its primary roles are to **maintain seminal pH and facilitate semen liquefaction**.

BENIGN VERSUS MALIGNANT PROSTATE EPITHELIAL METABOLISM

Feature	Benign	Malignant
Intracellular zinc	High	Dramatically reduced
Mitochondrial aconitase	Inhibited	Uninhibited
Fate of citrate	Secreted end product	Oxidised in the Krebs cycle → ATP
Energy efficiency	Inefficient	Restored

Concept — the prostate is built for secretion at the expense of its own metabolism. **Zinc blocks aconitase, so citrate is exported rather than burned.** Malignant transformation reverses this by **losing zinc and regaining efficient citrate oxidation**.

KEY POINTS

- Prostate anatomy has moved from **morphological landmarks to a cellular anatomy** that defines structure and provides a framework for function.
- Mouse and human prostate show **key cellular and anatomic differences**.
- **AR regulation** appears to be one of the most important "gatekeepers" of prostate development and physiology.
- Secreted prostate proteases have a **significant, dose-dependent effect on semen liquefaction**.

- Prostate secretions support sperm motility and survival. Some secreted factors are clinically valuable as **cancer biomarkers, imaging agents and therapeutic targets**, and potentially for male contraception.

PROSTATE DEVELOPMENT

Overview and Timeline

- **Why development is studied:** ductal morphogenesis **requires androgens** (a model of androgen action); it shows how the microenvironment directs epithelial fate (insight into **BPH, cancer and wound healing**); and **embryonic chemical exposure** has lasting effects on differentiation, proliferation, testosterone response, innervation and smooth muscle activity.
- **Origin:** the **urogenital sinus (UGS)**, a transient urethral widening near the bladder neck, formed by cloacal division (timeline table).
- **Incomplete cloacal division** (imperforate anus, urorectal fistula) is associated with **impaired prostate development**.
- Rodent models are valid because **rodent prostate development resembles the human**.
- The UGS is initially **ambisexual**: a sheath of **mesoderm-derived mesenchyme** around a tube of **endoderm-derived epithelium**. AR is widely expressed in both male and female UGS, and **both form prostate if exposed to sufficient androgen**.
- The **prostate is absent after castration or with an inactive AR** in rodents. It is **incomplete in males homozygous for a loss-of-function variant of SRD5A2** (steroid 5 α -reductase type 2).
- **Androgen is not rate limiting**. Testosterone acts on UGS mesenchyme soon after synthesis begins (about week 9), yet budding waits until week 10 and **cannot be accelerated in vitro** by physiological or supraphysiological androgen.
- **Epigenetic checkpoint:** the **AR gene is highly methylated in the primitive UGS**. Demethylation raises AR abundance and makes the UGS receptive to androgen-induced budding.

TIMELINE OF HUMAN PROSTATE DEVELOPMENT (WITH FIG. 148.5)

Gestational week	Stage	Event
6–7	UGS formation	Cloacal division into ventral genitourinary and dorsal anorectal tracts
6–9	Primitive UGS	Epithelial tube within a mesenchymal sheath; AR gene methylated
About 9	Instructive induction	Testosterone activates mesenchymal AR → andromedins instruct epithelium to become prostate
10	Bud formation	Solid epithelial cords invade mesenchyme (bidirectional signalling); bud pattern sets mature ductal pattern
11 onwards	Elongation, canalisation, branching	Elongation towards chemotactic factors (distal-tip proliferation, altered cell adhesion); canalisation from urethra outwards to bud tips; branching through puberty and beyond; differentiation into a secretory network

Concept — androgen is present by week 9, but buds wait until week 10. The **gate is receptor availability (AR demethylation)**, not hormone supply.

Molecular Control of Prostatic Bud Formation and Patterning

- Prostate development is initiated by functional AR in UGS mesenchyme, not in UGS epithelium (Lasnitzki and Mizuno, 1980).

- **Lasnitzki-Mizuno experiment:** UGS from control mice and from mice with a spontaneous AR mutation introducing a premature stop codon (truncated, nonfunctional protein) → enzymatic separation of mesenchyme and epithelium → recombination in different combinations → organ culture with testosterone → **buds only when the mesenchyme carried functional AR** (wild-type recombinants budded; mutant recombinants did not).
- **Induction** is the process by which one group of cells changes the behaviour of a nearby group. Holtzer (1968) described two modes, in the context of bone formation.

PERMISSIVE VERSUS INSTRUCTIVE INDUCTION (HOLTZER)

Feature	Permissive	Instructive
Responding tissue	Already holds the information; awaits permission	Lacks the information; must be programmed
Prostate scenario	AR signal merely allows epithelium to begin	Mesenchyme programmes epithelium to become prostatic, ductal, secretory
Applies to UGS mesenchyme	No	Yes

- **Cunha recombination:** fetal mouse UGS and adult bladder separated into epithelium and mesenchyme → recombined → grafted under the renal capsule of male mice → **UGS mesenchyme reprogrammed bladder epithelium into prostatic glandular epithelium: an instructive inducer.**
- **Specification:** UGS epithelial cells are specified once they **coexpress Nkx3-1, Edar, Wnt10b** and other factors.
- UGS mesenchyme **loses inductive capacity with age; McNeal hypothesised** that ageing stroma may **regain it**, initiating new glands and BPH.
- **Andromedin hypothesis:** mesenchymal AR drives synthesis of proteins (andromedins) secreted or presented to epithelium. **Three criteria:** (1) androgen-dependent or androgen-enhanced expression; (2) required for development; (3) ectopically enhances development or substitutes for androgen.

ANDROMEDIN CANDIDATES IN UGS STROMA

Factor	Androgen-induced in UGS stroma	Required for development	Ectopic protein enhances development
Wnt2 (wingless-type MMTV integration site family, member 2)	Yes	—	—
Secreted frizzled-related protein 2	Yes	—	—
Wnt inhibitory factor 1 (Wif1)	Yes	—	Yes
Fibroblast growth factor 10 (Fgf10)	Yes	Yes	Yes
Fibroblast growth factor 7 (Fgf7)	Yes	—	Yes

- The table tests the criteria. **No single factor has yet been shown to have all three andromedin properties.** Combinations may substitute for androgen in initiating development, with **major roles for retinoic acid and FGF10.**
- **Lifelong regulators:** stromal inducers and negative regulators shape morphogenesis throughout life; signals first found in fetal budding — FGF10, homeobox genes, hedgehog genes, retinoic acid signalling, bone morphogenetic proteins (BMPs), WNT5A — were **each later implicated in prostate disease.**
- **Patterning:** androgens stabilise buds in a **dorsoventral and left-right pattern** fixing mature gland position, partly through **β-catenin** (required by nearly all bud-forming tissues). β-catenin drives proliferation **cell-autonomously** and induces **inhibitors of budding in nearby epithelium**, setting **bud size and spacing.**

Prostatic Bud Elongation and Branching Morphogenesis

- Branching gives a **large surface area-to-volume ratio**, increasing capacity for fluid synthesis and storage.
- **Branching requires androgens** — the prostate **remains unbranched in humans with inactivating AR variants** and is **impaired by neonatal castration** — acting through **stromal, not epithelial, AR**, with **bidirectional paracrine signalling** using many budding pathways.

REGULATORS OF PROSTATIC BRANCHING MORPHOGENESIS

Class	Source → target	Factors
Stromal paracrine (stimulate or inhibit)	Stroma → epithelial receptors	Ephrin B1, FGF7, FGF10, transforming growth factor β 1, β 2 and β 3 (TGFB1–3), BMP4, BMP7, activin, Notch ligands
Epithelial cell-autonomous	Act within the epithelial cell	NKX3-1, TRP63, HOXA13, HOXD13
Epithelial paracrine	Epithelium → mesenchyme	Sonic hedgehog , which activates mesenchymal factors
Neutralising modulators	UGS epithelium or mesenchyme → bind activators or inhibitors	Noggin, heparan sulfate, secreted frizzled-related proteins

- The neutralising modulators **fine-tune growth-factor gradients** that control branching pattern and interval. The trap is **sonic hedgehog**, the one epithelial factor that acts in a **paracrine** manner, on the mesenchyme.

Concept — the mesenchyme instructs and the epithelium responds. AR is required in the stroma for both budding and branching, andromedins carry the message, and β -catenin-driven local inhibition sets bud spacing.

Environmental Factors Influence Prostate Morphogenesis

- Gestation is a window of vulnerability to environmental and dietary agents. **Developmental origins of health and disease (DOHaD) hypothesis:** early-life (including gestational) chemical exposures modify the risk of ageing diseases such as diabetes, cancer and hypertension; their effect on prostate disease is **poorly understood**.
- **Oestrogen:** androgens are the principal driver, but oestrogen receptors are expressed lifelong and are abundant in fetal stroma and epithelium, including progenitors. Oestrogens are **not required for budding or branching but may be required for patterning** (fewer branches in mice lacking oestrogen receptor 1 and 2 activity).
- Oestrogenic substances are **full or partial receptor agonists**; gestational or lactational exposure in rodents causes prostatic **inflammation, dysplasia, neoplasia**, even adenocarcinoma in susceptible animals.

ENVIRONMENTAL AGENTS AFFECTING PROSTATE DEVELOPMENT

Agent	Class	Exposure and species	Effect
Di(2-ethylhexyl) phthalate (DEHP)	Plasticiser in poly(vinyl chloride)	Gestational or lactational; rat	Reduced lobe weight; agenesis of anterior and ventral prostate
2,3,7,8-Tetrachlorodibenzo-p-dioxin (TCDD)	Persistent pollutant	Gestational; mouse and rat	Ventral prostate agenesis (mouse); transgenerational histopathology (rat); hyperinnervation (mouse)
Diethylstilboestrol (DES)	Synthetic oestrogen; potent oestrogen-receptor agonist	Gestational; mouse, rat, human	Dose-dependent reduction or increase in dorsal budding (rodent); squamous metaplasia and neoplasia in the postnatal prostate (human male)

Agent	Class	Exposure and species	Effect
Oestradiol benzoate, bisphenol A	Oestrogenic	Perinatal rat; human prostate organoids	Sustained changes in histone and DNA methylation
Vinclozolin	Fungicide	During gonadal sex determination; rat	Heritable epimutations over at least three generations ; altered prostate DNA methylation and gene expression
Glyphosate; methoxychlor	Herbicide; insecticide	During gonadal determination; rat	Transgenerational histopathology (glandular atrophy or hyperplasia)

THREE DEVELOPMENTAL-ORIGIN MECHANISMS LINKING EARLY EXPOSURE TO LATER PROSTATE DISEASE

Mechanism	Linked exposure	Consequence
Epigenetic imprinting of prostate progenitor cells	Oestrogenic chemicals	Persistent histone and DNA methylation changes
Germline epimutations	Vinclozolin, TCDD, glyphosate, methoxychlor	Transgenerational changes in prostate methylation and histopathology
Altered prostate innervation	TCDD	Hyperinnervation; hyperactive smooth muscle; urinary frequency

- **Innervation:** smooth muscle activity and the response to **α -adrenergic receptor blockers** in lower urinary tract dysfunction **vary widely between men**; the controlling autonomic neurons first innervate the prostate **during gestation**.
- **Sequence:** gestational TCDD → **increased artemin** (a neurotrophic factor) in UGS epithelium → **more noradrenergic fibres** in male fetal mice → hyperinnervation **persisting into adulthood** → thickened, hyperactive smooth muscle and **increased urinary frequency**.

Concept — a fetal exposure can reset the adult prostate through **three routes: the progenitor epigenome, the germline and the nerve supply**. The innervation route offers one explanation for the variable response to α -blockers.

KEY POINTS

- Prostatic **budding, branching and cytodifferentiation** are driven by signals from the stroma/mesenchyme.
- The embryonic period is **susceptible to external factors**, such as endocrine-disrupting and other chemicals, that can influence anatomic development and cell differentiation.